WEEK 6 – DATA SCIENCE AND VISUALIZATION

Normalization, Scaling, Categorical Encoding and Skewness

Project	Dataset	Syllabus Mapping
Data Analysis Project	Cleaned_Dataset.csv	Unit 3

1. Introduction

Week6 prepares data for later analysis through feature engineering. Numerical features are normalized or standardized, categorical values are converted into numeric indicators, and numerical skewness is examined. The original dataset is preserved while demonstration outputs are created separately.

2. Objectives

• Inspect dataset dimensions and data types.
• Identify numerical and categorical columns.
• Apply Min-Max normalization and Standard scaling.
• Encode selected categorical columns using one-hot encoding.
• Inspect skewness and demonstrate log1p transformation for suitable non-negative features. •
Save transformed outputs without overwriting the source dataset.

3. Dataset Used

Dataset:**Cleaned_Dataset.csv**. The Colab preview showed 156,944 rows and 53 columns, including Type, Days for shipping (real), Days for shipment (scheduled), Benefit per order, Sales per customer, Delivery Status, Late_delivery_risk, Category Name, Customer City and Product Price. Recheck the shape when running the notebook.

4. Tools and Technologies

Python,Pandas,NumPy,Matplotlib,Seaborn and Scikit-learn.

5. Load Dataset and Inspect

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('/content/Cleaned_Dataset.csv')
print("Dataset shape:", df.shape)
display(df.head())
print(df.dtypes.value_counts())
```

6. Identify Feature Types

```
target = 'Late_delivery_risk'
numeric_cols = df.select_dtypes(include=np.number).columns.tolist()
feature_numeric_cols = [c for c in numeric_cols if c != target]
categorical_cols = df.select_dtypes(include=['object', 'category']).columns.tolist()

print("Numerical input features:", len(feature_numeric_cols))
print("Categorical features:", len(categorical_cols))
print("Categorical examples:", categorical_cols[:10])
```

7. Min-Max Normalization

Min-Maxscalingmapsvaluestoarange from 0 to 1. A sample of up to five non-constant numerical features is used for a clear demonstration.

```
from sklearn.preprocessing import MinMaxScaler, StandardScaler

scale_cols = [c for c in feature_numeric_cols if df[c].nunique(dropna=True) > 1][:5]
ifscale_cols:
    sample = df[scale_cols].head(5000).copy()
    sample = sample.fillna(sample.median(numeric_only=True))
    normalized = pd.DataFrame(
        MinMaxScaler().fit_transform(sample), columns=scale_cols
    )
```

8. Standard Scaling

Standard scalingtransforms each selected feature to approximately zero mean and unit standard deviation. It is useful when features have different units or ranges.

```
if scale_cols:
    sample = df[scale_cols].head(5000).copy()
    sample = sample.fillna(sample.median(numeric_only=True))
    standardized = pd.DataFrame(
        StandardScaler().fit_transform(sample), columns=scale_cols
    )
    display(standardized.head())
    print("Means:", standardized.mean().round(3).to_dict())
    print("Standard deviations:", standardized.std(ddof=0).round(3).to_dict())
```

9. Categorical Encoding

One-hot encoding convertscategorylabels into 0/1 indicator columns. This example chooses up to three categorical columns with 2–20 unique values, avoiding very high-cardinality text columns.

```
from sklearn.preprocessing import OneHotEncoder

candidate_cats = [c for c in categorical_cols if 2 <= df[c].nunique(dropna=True) <= 20][:3]
if candidate_cats:
    cat_sample = df[candidate_cats].head(5000).fillna("Missing").astype(str)
    encoder = OneHotEncoder(handle_unknown='ignore', sparse_output=False)
    encoded_array = encoder.fit_transform(cat_sample)
    encoded_df = pd.DataFrame(
        encoded_array, columns=encoder.get_feature_names_out(candidate_cats)
    )
    print("Columns  encoded:",  candidate_cats)
    print("Encoded  shape:",  encoded_df.shape)
    display(encoded_df.head())
else:
    print("No low-cardinality categorical columns found.")
```

10. Skewness Analysis

Skewness measures distribution asymmetry. Large positive values often indicate a long right tail. The code ranks numerical columns by absolute skewness and plots the most skewed column.

```
skewness = df[feature_numeric_cols].skew(numeric_only=True).dropna()
skewness = skewness.sort_values(key=lambda s: s.abs(), ascending=False)
display(skewness.head(10).to_frame('Skewness'))

if len(skewness):
    col = skewness.index[0]
    values = df[col].dropna()
    values = values.sample(min(5000, len(values)), random_state=42)
    plt.figure(figsize=(8, 4))
    sns.histplot(values, kde=True)
    plt.title(f'Distribution of {col}')
    plt.tight_layout()
    plt.show()
```

11. Demonstrate Skewness Correction

Alog1p transformation is demonstratedonly fornon-negative columns with strong positive skew. The transformed feature is stored in a new column, leaving the original column unchanged.

```
positive_skew_cols = [
    col for col in skewness.index
    if skewness[col] > 1 and df[col].dropna().ge(0).all()
][:3]

for col in positive_skew_cols:
    new_col = f'{col}_log1p'
    df[new_col] = np.log1p(df[col])
    print(col, "before:", round(df[col].skew(), 3),
          "after:", round(df[new_col].skew(), 3))
```

12. Save Outputs

```
if positive_skew_cols:
    df.to_csv('Week6_feature_engineered_dataset.csv', index=False)
    print("Saved: Week6_feature_engineered_dataset.csv")

if scale_cols:
    normalized.to_csv('Week6_minmax_sample.csv', index=False)
    standardized.to_csv('Week6_standard_scaled_sample.csv', index=False)
    print("Savedscaled  sample files.")
```

13. Results to Record

Fillintheseresultsafterrunningthenotebook. Values depend on the actual dataset and should not be guessed.

Item	Record from output
Dataset dimensions	Printed shape (rows × columns)
Numerical / categorical feature counts	Counts from Section 6
Min-Max range	Minimum and maximum values
Standard-scaled means	Means should be close to 0
Encoded columns	Names printed in Section 9
Skewness before and after	Values printed in Section 11

14. Conclusion

Week 6demonstrates the key feature-engineering steps: scaling numerical data, encoding selected categorical variables and examining skewness. The methods help make features more suitable for analysis while keeping the original dataset available. Final findings should be reported from the actual notebook output.

15. Project Status

Week6:FeatureEngineering — notebook steps prepared; run the cells and record the observed results before submission.